03 June

Database:

- An organazed collection of data - typically spliting tables that relate to each other.
- An Organized collection of raw facts, Stored ellectronically in a computer, allowing users and applications to effe store, manage, searc, and retrieve large volumes of information.
  - Any collection of related information

Types of databas(s)

(records) (fields)

1. **Relational (SQL) Database:** Stores data into rows and columns Tables can be linked together using specific relationships This use SQL to define, update, and query data. E.g MySQ Postgre, and Oracle

2. **NOSQL Database:** Build to handle unstructured or varied data and are highly scalable. Instete of using tables, may store data as documents, key value pairs, or graphs. Pictures.

3. **Vector Database:** It stores data as vector embedding, whic Is heavly used to power generative AI modles and Sematic

WHY ARE THEY IMPORTANT

Unlike Spreadsheet: which can crush under massive loads and requre duplicate data. Databases enforce rules (Schemas) to prevent errors and ensure data accurancy. (they are unseen engines powering everyday digital experiences, such as banking transactions, Social media feeds, e-commerce inventory manag

1. Create GitHub (for project submissions and portfolio develop

TO DRIVE BUS. DESCIIONS.

- TIME FACTOR -

03 - Introduction to database.

- data - how - Works - Systems involved?

1. Data?! Collection of facts! or Information collection Avable to answer bissines questoons.

-> It can come in different formats (voice) video / is everywhere

2 main things in data

i) Structured data: refers to data stored in a table w row & columns Consistent formant

ii) Unstructured data; It has no specific formart: Inconsistant format.
e.g music. Potcast. Social medi post. Picture. Video. preaches, PDF documents. Adverts. web side. (Youtube brighteam)

e↓

	Col.	Co 2	Co 3	Co.4	Co.5
R1					
R2					
R3					

(i) e.g - Spreadsheet (Excel file)
- member form. tinder

Name:

Surname:

Age:

Location:

(front-end)

unique key (identifer)

- Back end -

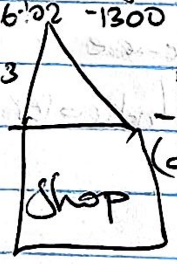
id	name	Surname	Age	lo
1.	Nthabi	Rake	33	[illegible]

Unique id (Order #) to identify a different transaction. to link in a new data. - to data driven descions in a bussiow Pos-system.

Order-table

order-id	Product	Rave Price	datetime
101	Str 2	40	2026-06-02 -1300
110	Ice-Cream	10	206-06-3
204	All Stars	80	

Pound sale



pos system
-> connected to a Server (database) - Server Rooms -
- cold
- no loadsheeding

Intraction - Smart Shoper! on database
=> back at the server (data Room)
-> look alike content based on what you bought! vid medias

Data Collection: For business descions based on productions and sales give us information of customers to drive descion based on high possible fishing! target market. time. media.

- Changes happen based on (datetime)
  - also on hyper date

cross sell - + icecream
up sell - wise2 - 40 -> 3P R 60

- BASed ON previous data
- Study trends and patterns
- data-driven descions.

dateTime, Location, Production, Revenue
xx xx xx

Body lotion ① 40 ② 60 upsell

Cross-sell lotion ① ~~40~~ 100
tShowergel + 5

body lotion - 60
shower gell - 50

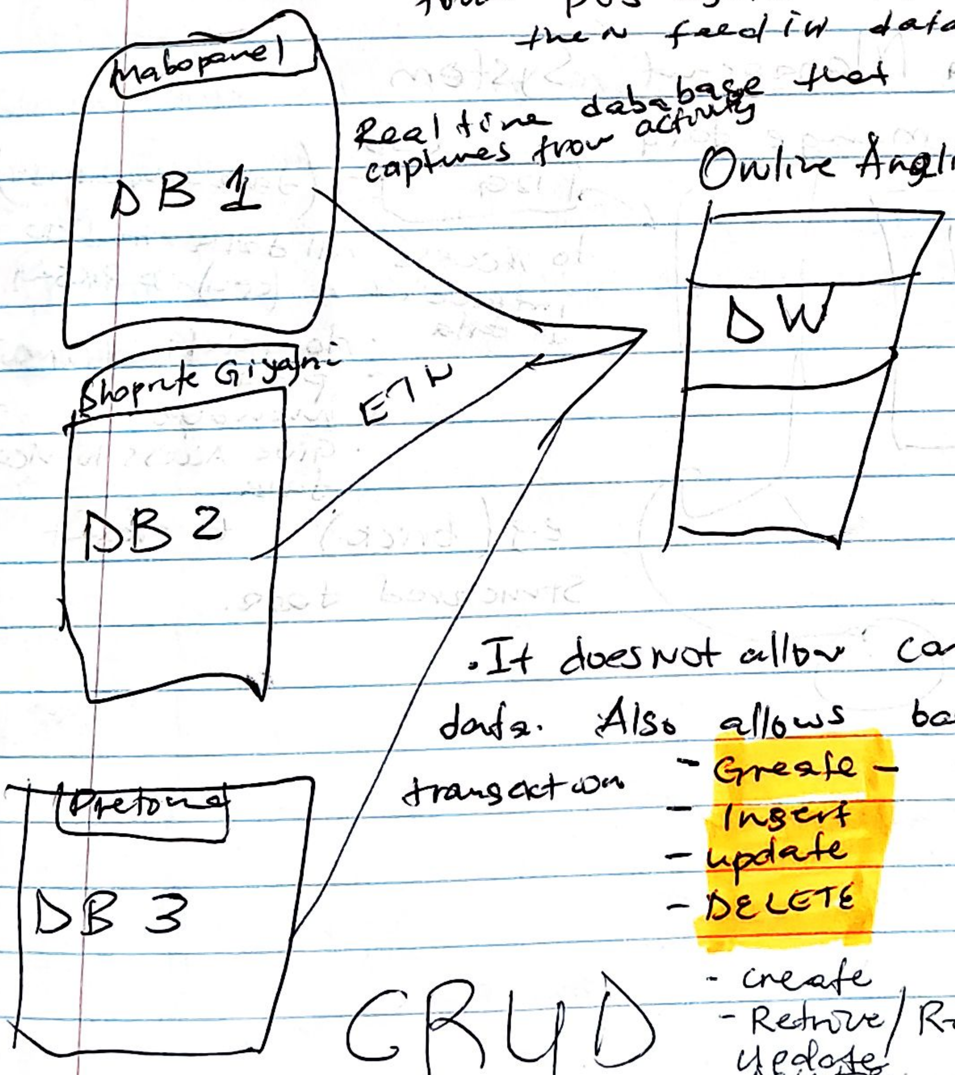
Voice bundle + data bundles

(bootcamp - data Visualization) - Free
* What is data Science*

04 June

Data Flow: from different places -> data warehouse

e.g Shoprite. OLTP - OnLine trasactions Processing data
from pos system stored in Database (OLDB)
then feed in datawarehouse.



Real time database that captures from activity

Online Anglitical Processing (OLAP)
or warehouse.

- how does it move
move via ETL
Extract Tranform Load
- used by Analysts & Scientist

. It does not allow complex querys or Analyze data. Also allows basic (SQL) deletes transaction

- Create -
- Insert
- update
- DELETE

CRUD
- create
- Retrive/Read
Update
Delete

What You do in the DB

wb

ETL - Extract, Transfom, Load: Pipeline to ensure data in the data base the warehouse designed by ~~Data~~ Data Engeemes. ~ Data Architeme - plan on the Collected data. So data Analyst (Scientist) can predict maps the future of the store.

Serve Roones - (Source Systems) from Pos.

DataWareHouse: Ready Home for Company's data! It Stores Clean, Organize historical data from many Sources. For Analytical process.

B. I - upsell / cross sell

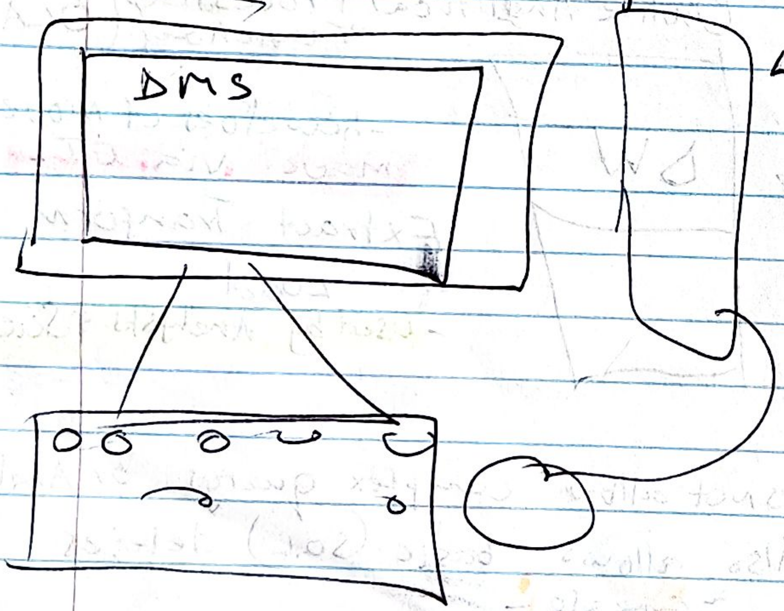
MD - buy 1 get One free.

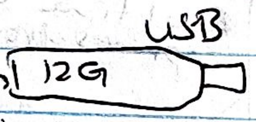
ELT - Extract - Load, Transform ⟶ Load data into warehouse First!

* Data Ingestion *

~~M~~ DMS Data Managment System

Computer System to mange data.





(data warehouse)

to Access All data had to interface a A (cpu) to present all data
- to delete
- play
- manage
- Give Access to view data.

e.g (brick) to present Structured data.

-(Contain descriptive (information